

Name: _____

Phys 12

Worksheet 6.02 Electric Field

- 1) What is the electric field strength 0.750 m from an 8.00 μC charged object? (1.28x10⁵ N/C)

- 2) Calculate the gravitational field strength on the surface of Mars. Mars has a radius of 3.43x10⁶ m and a mass of 6.37x10²³ kg. (3.61 N/kg)

- 3) At a point a short distance from a 4.60x10⁻⁶ C charged object, there is an electric field strength of 2.75x10⁵ N/C. What is the distance to the charged object producing this field? (0.388 m)

- 4) If an alpha particle experiences an electric force of 0.250 N at a point in space, what electric force would a proton experience at the same point? (0.125 N)

- 5) What is the electric field strength at a point in space where a 5.20x10⁻⁶ C charged object experiences an electric force of 7.11x10⁻³ N? (1370 N/C)

- 6) What is the initial acceleration of an alpha particle when it is placed at a point in space where the electric field strength is 7.60x10⁴ N/C? (3.66x10¹² m/s²)

- 7) Calculate the electric field strength midway between a 4.50 μC charged object and a - 4.50 μC charged object if the two charges are 50 cm apart. (1.30x10⁶ N/C)

- 8) Calculate the electric field strength midway between a 3.0 μC charged object and a 6.0 μC object if they are 0.80 m apart. (1.7x10⁵ N/C)

- 9) Calculate the electric field strength midway between two 3.0 μC objects if they are 90 cm apart. (0 N/C)